

Here's a serious enterprise-style BRD for your project. This is written more like something used in real AI product planning rather than a college assignment.

Business Requirements Document (BRD)

Project Title

ReguMind — Enterprise Regulatory Intelligence & Verified RAG Platform

Version

1.0

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Document Type

Business Requirements Document (BRD)

Project Category

Enterprise Artificial Intelligence Platform
Retrieval-Augmented Generation (RAG) System
Regulatory Intelligence Infrastructure

1. Executive Summary

Organizations across healthcare, finance, legal, insurance, compliance, and government sectors manage massive volumes of regulatory and operational documents. Employees spend significant time manually searching through policies, filings, legal notices, manuals, compliance guidelines, and regulatory updates to locate accurate information.

Traditional keyword-based enterprise search systems fail to:

- understand natural language queries
- retrieve semantically relevant information
- validate answer correctness
- provide grounded evidence
- scale efficiently across large document repositories

ReguMind is an enterprise-grade AI-powered Regulatory Intelligence Platform designed to ingest, process, index, retrieve, validate, and generate evidence-grounded answers from large-scale regulatory and enterprise document repositories exceeding 100,000+ documents.

The platform leverages Retrieval-Augmented Generation (RAG), hybrid retrieval, vector search, reranking, hallucination detection, multi-agent orchestration, and real-time observability to provide trustworthy AI-assisted decision support.

The system prioritizes:

- reliability
 - traceability
 - explainability
 - scalability
 - compliance
 - enterprise trustworthiness
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2. Business Problem Statement

Enterprises face significant operational inefficiencies due to fragmented and rapidly evolving regulatory information spread across:

- policy documents
- legal filings
- healthcare regulations
- government publications
- operational manuals
- audit documents
- compliance frameworks

- financial disclosures

Current challenges include:

- excessive manual searching
- inconsistent interpretation of policies
- delayed decision-making
- inability to trace information sources
- regulatory risk exposure
- knowledge silos
- outdated enterprise search infrastructure
- hallucinated AI responses from generic chat systems

Organizations require an intelligent system capable of:

- understanding complex queries
 - retrieving precise evidence
 - generating grounded answers
 - detecting unsupported responses
 - scaling to massive document repositories
 - maintaining enterprise-grade observability and auditability
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3. Business Objectives

The primary business objectives of ReguMind are:

3.1 Operational Efficiency

Reduce time spent searching enterprise documents by at least 70%.

3.2 Regulatory Accuracy

Provide evidence-grounded responses with source traceability.

3.3 Decision Support

Enable faster and more informed compliance and operational decisions.

3.4 Enterprise Scalability

Support ingestion and retrieval across 100,000+ documents.

3.5 Hallucination Reduction

Minimize unsupported AI-generated responses using verification workflows.

3.6 Knowledge Centralization

Create a unified intelligent knowledge platform for enterprise regulatory data.

3.7 Observability & Monitoring

Provide full monitoring for retrieval quality, response confidence, latency, and hallucination metrics.

4. Scope

4.1 In Scope

The platform shall include:

Document Ingestion

- PDF ingestion
- HTML ingestion
- XML ingestion
- DOCX ingestion
- API-based ingestion
- Batch ingestion pipelines

Data Sources

- Federal Register
- CMS Guidelines
- SEC EDGAR Filings
- Regulations.gov

- CourtListener
- Internal enterprise documents

NLP Processing

- OCR support
- semantic chunking
- metadata extraction
- embedding generation

Retrieval System

- vector search
- hybrid retrieval
- BM25 search
- metadata filtering
- reranking

AI Capabilities

- question answering
- summarization
- citation generation
- compliance reasoning
- answer validation
- hallucination detection

Enterprise Features

- confidence scoring
- source attribution
- audit logging
- monitoring dashboard
- analytics dashboard
- API access

Infrastructure

- scalable vector database
- cloud-ready deployment
- Docker support

- asynchronous processing
 - caching layer
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4.2 Out of Scope

The following are excluded from Phase 1:

- autonomous legal advice
 - autonomous medical diagnosis
 - fully automated regulatory decision-making
 - replacing human compliance officers
 - multilingual translation support
 - real-time voice interaction
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5. Stakeholders

Stakeholder	Role
Compliance Officers	Regulatory validation
Legal Teams	Legal intelligence retrieval
Healthcare Analysts	Healthcare policy analysis
Risk Management Teams	Risk assessment support
Enterprise Employees	Knowledge retrieval
AI Engineering Team	System development
DevOps Team	Infrastructure & deployment
Security Team	Access control & auditing
Executive Leadership	Strategic oversight

6. Functional Requirements

6.1 Document Ingestion Module

FR-1

The system shall ingest documents from multiple formats including PDF, DOCX, XML, HTML, and TXT.

FR-2

The system shall support bulk ingestion of 100,000+ documents.

FR-3

The system shall support scheduled ingestion pipelines.

FR-4

The system shall extract metadata including:

- title
 - source
 - publication date
 - regulation type
 - agency
 - jurisdiction
 - document category
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6.2 Document Processing Module

FR-5

The system shall perform semantic chunking using heading-aware segmentation.

FR-6

The system shall generate embeddings for all chunks.

FR-7

The system shall maintain document-to-chunk traceability.

FR-8

The system shall detect duplicate documents.

6.3 Retrieval Engine

FR-9

The system shall support vector similarity search.

FR-10

The system shall support BM25 keyword retrieval.

FR-11

The system shall implement hybrid retrieval combining semantic and keyword search.

FR-12

The system shall support metadata filtering.

FR-13

The system shall rerank retrieved results.

6.4 AI Reasoning Layer

FR-14

The system shall generate answers grounded only in retrieved evidence.

FR-15

The system shall generate source citations.

FR-16

The system shall refuse unsupported responses.

FR-17

The system shall provide confidence scores.

6.5 Hallucination Detection Layer

FR-18

The system shall validate generated answers against retrieved evidence.

FR-19

The system shall flag unsupported claims.

FR-20

The system shall maintain hallucination metrics.

6.6 Multi-Agent Workflow

FR-21

The system shall support agent orchestration.

Planned Agents:

- Query Understanding Agent
 - Retrieval Agent
 - Validation Agent
 - Hallucination Detection Agent
 - Summarization Agent
 - Citation Agent
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6.7 Dashboard & Monitoring

FR-22

The system shall provide monitoring dashboards for:

- latency
- query volume
- hallucination rate
- confidence distribution
- retrieval quality

FR-23

The system shall provide analytics dashboards.

6.8 Security & Compliance

FR-24

The system shall implement role-based access control.

FR-25

The system shall maintain audit logs.

FR-26

The system shall support encrypted storage.

7. Non-Functional Requirements

Requirement	Description
Scalability	Support 100k+ documents
Availability	99.5% uptime target
Performance	<5 second average response time
Reliability	High-confidence grounded responses
Security	Encrypted storage and secure APIs
Observability	Full logging and monitoring
Maintainability	Modular microservice architecture
Extensibility	Support future data sources

8. System Architecture Overview

The architecture shall contain:

Data Layer

- Raw document storage

- Metadata database
- Vector database

Processing Layer

- OCR pipeline
- chunking engine
- embedding pipeline

Retrieval Layer

- vector retrieval
- keyword retrieval
- reranking

AI Layer

- LLM inference
- agent orchestration
- hallucination detection

API Layer

- FastAPI services
- authentication
- rate limiting

Frontend Layer

- enterprise dashboard
- query interface
- analytics panel

Monitoring Layer

- logs
 - metrics
 - alerts
 - observability tools
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9. Technology Stack

Layer	Technology
Programming Language	Python
Backend Framework	FastAPI
AI Orchestration	LangChain / LangGraph
Vector Database	Qdrant
Metadata Database	PostgreSQL
Cache Layer	Redis
Embeddings	BGE Large / OpenAI
LLM	GPT-4 / Claude / Llama 3
Monitoring	Prometheus + Grafana
Containerization	Docker
Cloud	AWS
Frontend	Streamlit / React

10. Data Volume Expectations

Metric	Target
Documents	100,000+
Chunks	5M+
Daily Queries	10,000+
Average Chunk Size	500-1000 tokens
Metadata Records	100,000+

11. Risks & Mitigation

Risk	Mitigation
Hallucinated answers	Validation pipeline
Poor retrieval quality	Hybrid search + reranking
Large-scale indexing delays	Async ingestion
Regulatory updates	Scheduled sync pipelines
Duplicate documents	Deduplication layer
Slow inference	Caching + optimized retrieval

12. Success Metrics

The project shall be considered successful if:

- Retrieval accuracy exceeds 85%
 - Hallucination rate remains below 5%
 - Average response latency remains below 5 seconds
 - Citation coverage exceeds 95%
 - System scales to 100k+ documents
 - User trust/confidence scores improve significantly
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13. Future Enhancements

Potential future capabilities include:

- multilingual support
- voice query interface
- graph-based knowledge reasoning
- fine-tuned domain LLMs
- enterprise SSO integration
- document change tracking
- regulatory alerting system

- AI-powered risk scoring
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14. Conclusion

ReguMind aims to redefine enterprise regulatory intelligence by combining scalable Retrieval-Augmented Generation architecture with evidence-grounded reasoning, hallucination detection, observability, and enterprise-grade trust mechanisms.

The platform is designed not merely as a chatbot, but as a scalable AI knowledge infrastructure capable of supporting high-stakes enterprise decision-making across healthcare, finance, legal, compliance, and government domains.